FILE 8: questionnaire_mapping_and_scales.pdf

Description: Mapping of questionnaire sections to research variables, hypotheses and research questions for "Context-Aware Blockchain Adoption for Real Estate Fraud Mitigation in Ghana's Dual Land Tenure System."

Scale Definitions

- All Likert items use a 5-point scale.
- Agreement items: 1 = strongly disagree; 2 = disagree; 3 = neutral; 4 = agree; 5 = strongly agree.
- Perception/level items: 1 = very low; 2 = low; 3 = moderate; 4 = high; 5 = very high.
- Willingness items: 1 = very unwilling; 2 = unwilling; 3 = neutral; 4 = willing; 5 = very willing.

Mapping to Research Objectives and Hypotheses

Questionnaire Section	Research Variable / Construct	Hypothesis(es) Tested	Research Question(s)
Section A: Demographics and professional background	Stakeholder category, location, age, experience, prior BT training, BT literacy	Controls; H3	RQ1, RQ2
Section B: Blockchain literacy	BT_literacy_selfrated; comprehension_pass	H3	RQ1
Section C: Perceived blockchain capabilities	BT_immutability; BT_transparency; BT_automation; BT_capabilities	H1, H2, H3	RQ1
Section D: Institutional trust	IT_score (Lands Commission, courts, banks, customary authorities)	H1	RQ1, RQ2
Section E: Regulatory clarity	regulatory_clarity	H2	RQ1, RQ2
Section F: Oracle concerns	oracle_concerns	H4	RQ2

Questionnaire Section	Research Variable / Construct	Hypothesis(es) Tested	Research Question(s)
Section G: Fraud/corruption reduction and adoption intentions	perceived_fraud_reduction; perceived_corruption_reduction; adoption_intention	H1–H4	RQ1, RQ2
Section H: Open-ended responses	Qualitative themes on customary integration, oracle governance, legal framework, phased implementation	H2, H4	RQ1, RQ2

Variable Mapping

Variable	Questionnaire Items	Scale
BT_literacy_selfrated	B1–B6 plus comprehension check	1–5 Likert; pass/fail
BT_immutability	C1–C5	1–5 Likert
BT_transparency	C6–C10	1–5 Likert
BT_automation	C11–C15	1–5 Likert
IT_score	D1–D12	1–5 Likert
regulatory_clarity	E1–E4	1–5 Likert
oracle_concerns	F1–F6	1–5 Likert
perceived_fraud_reduction	G1–G2	1–5 Likert
perceived_corruption_reduction	G3	1–5 Likert
adoption_intention	G4–G6	1–5 Likert

Variable	Questionnaire Items	Scale
transaction_type	Scenario/judgement item	Statutory / Customary
prior_BT_training	A5	Yes / No
experience_years	A4	Ordered band

Construct-to-Hypothesis Mapping

Hypothesis	Independent Variable	Moderator	Dependent Variable
H1	BT_immutability	Institutional trust (inverted U)	perceived_fraud_reduction
H2	BT_automation	regulatory_clarity	perceived_corruption_reduction
H3	BT_transparency	BT_literacy	adoption_intention
H4	oracle_concerns	transaction_type	adoption_intention

Notes:

- The questionnaire was pre-tested with 30 stakeholders in Accra.
- Cronbach's α exceeded 0.70 for all constructs; reported values: BT literacy 0.87; BT capabilities 0.91; institutional trust 0.88; fraud vulnerability 0.89; adoption intentions 0.85; overall 0.93.
- A standardised 15-minute literacy preamble with five multiple-choice comprehension questions preceded the main survey.
- Prototype participants completed a modified version after the Hyperledger Fabric demonstration.